MIL-C-60861(MU)
6 December 1971
SUPERSEDING
FA-PD-MI-2453, Rev.1
5 May 1961

MILITARY SPECIFICATION
CARTRIDGE, DELAY, M128 ASSEMBLY

1. SCOPE

1.1 This specification covers Cartridge, Delay, M128 Assembly, for use in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-C-60539	- Coating, Anodic Conventional for Aluminum and Aluminum Alloys
MS28741	- Hose Assembly, Detachable End Fitting, Medium Pressure

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-413	- Visual Inspection Guide for Rubber O-rings

DRAWINGS

Munitions Command

D10520335	- Cartridge, Delay, M128 Assembly

FSC 1377

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DRAWINGS (Cont'd)

C8864168 - Carton, Inner, Packing, Ammunition for Cartridge, Delay, M174, M84, M85, M86, M87, M88, M89, M165, XM128, XM148, XM159 or M175
C9214156 - Box, Packing, Ammunition for Cartridge, Delay, M128
IEL10520335 - Inspection Equipment List for Cartridge, Delay, M128 Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ASA B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI) 1430 Broadway, New York, N.Y. 10018.)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall conform to the applicable drawings and referenced specifications.

3.2 Protective finish.-Protective finish shall be in accordance with MIL-C-60539.

3.3 Vibration.-The assembled cartridge shall withstand vibration as prescribed in MIL-E-5272, Procedure XII, without evidence of any disintegration of the delay charge, primer, igniter or propellant or derangement of components.

3.4 Primer sensitivity.-The primer shall fire when drop tested at an energy level of 26 inch-ounces in a sectioned cartridge, void of delay, igniter and propellant charges.

3.5 Functional

3.5.1 The cartridge shall not fail to fire.

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3.5.2 Ballistics.-The cartridge when tested in an M52 or M54 Initiator shall comply with the requirements specified in Table I.

Table I

Temperature	Pressure (psi) 1/ Min.	Max.	Delay Seconds (see 6.2)
-65° to -70° F	500	1500	6.0 Max.
65° to 75° F	575	1500	5.0± .30
160° to 165° F	625	1500	4.0 Min.

1/ Pressure values shall be produced in a 1 cubic inch pressure station at the end of 15 feet of aircraft hose, medium pressure, size no. 4.

3.6 Workmanship.-Workmanship shall be in accordance with the requirements of the applicable drawings, referenced specification and the following:

3.6.1 Metal defects.-The metal part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.6.2 Burr.-The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.6.3 Foreign matter.-The part or assembly shall not contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550.

4.2 First article sample.

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4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of: 50 cartridges for primer sensitivity, 5 for Sealant tests (to be selected from primer sensitivity sample) 30 cartridges for ballistic tests, 12 for Vibration test (to be selected from ballistic sample) and 10 sets of unassembled components (including explosive components) for dimensional and visual examination and tests. The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and tests.-After examination and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Ballistics.-The sample for ballistic tests shall be temperature conditioned and conform to requirements as indicated in Table I and II.

Table II

Temperature Degrees (^{o}) Fahrenheit (F)	Vibrated	Not Vibrated
-65 to -70	4	6
65 to 75	4	6
160 to 165	4	6

4.2.1.3 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.-An ammunition lot shall consist of an identifiable group of metal parts assemblies consisting of:

a. Components produced by a homogeneous process.
b. Propellant from not more than one lot.
c. Primers from not more than one lot.
d. Igniters from not more than one lot.
e. Delay elements from not more than one lot.

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4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables III and IV. For an individual defect (single characteristic) the criteria in Table III shall be used. For total defects within a major or minor class the criteria in Table IV shall be used. All non-conforming material shall be rejected. The roughness comparison specimens prescribed in ASA B46.1 shall be used as a basis of comparison for surface roughness determination. Inspection for rubber o-rings shall be in accordance with MIL-STD-413. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

Table III

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination	-	-	-	-
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

Table IV

Sample Size for Major Defects Number of defects in class	Total allowable defects per sample size			
	110	150	225	300
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
over 43	6	8	11	14

Sample Size for Major Defects Number of defects in class	110	150	225	300
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
over 48	8	11	17	20

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4.3.2.1 Delay Element M27, Dwg. D10520336, covering a detail of Dwg. D10520335

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
2. Mixed ammunition	Visual
3. Flash hole missing or obstructed	Visual
4. Marking missing, incorrect or illegible	Visual
5. Delay charge not visible through flash hole	Visual
MAJOR	
101. Protective finish inadequate or defective	Visual
102. Foreign matter	Visual
MINOR	
None defined	

4.3.2.2 Ring, Retaining, Dwg. 8597235, covering a detail of Dwg. D10520335

CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Concentricity of inside diameter with outside diameter	Gage
102. Diameter inside, max.	Gage
103. Diameter outside, min.	Gage
104. Thickness	Gage
105. Protective finish inadequate or defective	Visual
106. Burr	Visual
107. Foreign matter	Visual
MINOR	
201. Diameter, inside, min.	Gage
202. Diameter, outside, max.	Gage
203. Total length	Gage
204. Radius missing or incorrect	Visual
205. Surface finish improper	Visual

4.3.2.3 Case, Cartridge, Dwg. C10519996 covering a detail of Dwg. D10520335

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4.3.2.3 (Cont'd)

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Concentricity of head with body	Gage
102. Straightness of base	Gage
103. Diameter of head, min.	Gage
104. Thickness of wall, min.	Gage
105. Thickness of base, min.	Gage
106. Protective finish inadequate or defective	Visual
107. Burr or sharp corner	Visual
108. Foreign matter	Visual
MINOR	
201. Diameter of head, max.	Gage
202. Diameter of body	Gage
203. Total length	Gage
204. Thickness of wall, max.	Gage
205. Thickness of base, max.	Gage
206. Location of datum diameter	Gage
207. Radius missing or incorrect	Visual
208. Surface finish improper	Visual

4.3.2.4 Cartridge, M128 Assembly, Dwg. D10520335

CRITICAL	
1. Any part missing	X-ray/& Visual
2. Improper assembly	X-ray/& Visual
3. Mixed ammunition	Visual
4. Metal defective	Visual
5. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Concentricity of head with body	Gage
102. Diameter of cartridge head recess, min.	Gage
103. Depth of cartridge head recess	Gage
104. Diameter of body, max.	Gage

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4.3.2.4 (Cont'd)

Defect	Method of Inspection
MAJOR (Cont'd)	
105. Diameter of cartridge head, max.	Gage
106. Location of datum diameter	Gage
107. Cartridge profile failure	Gage
108. Firing pin or retaining cup above flush with face of head	Gage
109. Foreign matter	Visual
MINOR	
201. Total length	Gage
202. Fold, wrinkle, dent, buckle, bulge	Visual
203. Varnish inadequate, or missing	Visual

1/ The cartridge shall be x-rayed with the long axis of the cartridge perpendicular to the rays of the x-ray machine. The x-ray negatives, properly identified, shall be forwarded to the procuring agency.

4.3.2.5 Carton, Inner, Packing, Ammunition for Cartridge Delay M128 Assembly, Dwg. C8864168

CRITICAL

None defined

MAJOR

None defined

MINOR

201. Contents loose	Manual & Visual
202. Quantity of contents incorrect	Visual
203. Separator or filler missing	Visual
204. Heat sealer incomplete, loose or distorted	Visual
205. Cut or damage through impregnated box	Visual
206. Marking missing, incorrect or illegible	Visual

4.3.2.6 Box Packing for Cartridge Delay, M128, Assembly Dwg. C9214156.

CRITICAL

None defined

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4.3.2.6 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Contents loose	Manual & Visual
202. Hardware missing, loose, broken or improperly engaged	Visual
203. Strapping missing, loose, broken or improperly assembled	Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of cartons incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.3.3 Inspection tesing.-The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Protective finish.-Unless otherwise specified, sampling plans and procedures in determination of the acceptability of coated parts and articles submitted by a supplier shall be in accordance with provisions set forth in MIL-STD-105.

4.3.3.2 Hardness.-Fifty units shall be selected for this test. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot.

4.3.3.3 Airtightness.-Each unit shall be subjected to the airtightness test. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot.

4.3.3.4 Vibration.-Twelve of the 30 assembled, x-rayed cartridges of the initial production sample, selected for ballistic tests and properly identified shall be subjected to vibration as in 3.3. After the vibration test, the 12 vibrated cartridges shall be x-rayed as directed in Note 1/ of 4.3.2.4. Evidence of any disintegration of the primer, propellant, igniter, or delay charge or derangement of any component shall be cause for rejection of the lot. When a change occurs in the lot of primers, propellant or igniters after the initiation of production, 12 assembled cartridges from the firt production lot shall be tested in accordance with this paragraph.

4.3.3.5 Primer sensitivity.-A sample of 50 assembled cartridges from the initial production sample and 25 cartridges from each subsequent lot shall be tested for sensitivity. Failure of any primer to comply with the requirements of 3.4 shall be cause for rejection of the lot.

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4.3.3.6 Sealant application.-A sample of 5 cartridges (consumed in the primer sensitivity test) shall be disassembled and examined for improper sealant application. Failure of any cartridge to comply with the drawing requirements shall be cause for rejection of the lot.

4.3.3.7.1 Ballistics.-A sample of 30 assembled cartridges from each production lot shall be ballistically tested after temperature conditioning in accordance with Table V and 4.4.5.1. Failure of any cartridge to comply with the requirements of 3.5 shall be cause for rejection of the lot.

Table V

Temperature Degrees(°) Fahrenheit(F)	Sample
-65 to -70	10
65 to 75	10
160 to 165	10

4.3.4 Inspection equipment.-The examination and tests shall be performed using equipment listed on IEL 10520335.

4.4 Test methods and procedures.

4.4.1 Protective finish.-Test procedures for protective finish shall be in accordance with MIL-C-60539.

4.4.2 Hardness.-The hardness test shall be performed on all heat-treated components prior to the application of protective finish. A hardness testing machine shall be used to determine compliance with the drawing requirements.

4.4.3 Airtightness.-The method for determining airtightness shall be by use of the equipment listed on IEL 10520335.

4.4.4 Vibration.-The method and procedure for performing the vibration test shall be in accordance with MIL-E-5272, Procedure XII.

4.4.5 Primer sensitivity.-Remove and discard the charged delay body and all powder charges from the test cartridges by sectioning the cartridge case around its periphery 1 inch maximum above the base. Drop test 50 sectioned cartridges from the initial production sample and 25 sectioned cartridges from each production lot at 13 inches, using a steel ball 15/16 inches in diameter, weighing 1.94 ounces plus or minus 0.02 ounces in a primer sensitivity machine with a cartridge holder, approved by the procuring activity.

4.4.6 Sealant application.-The cartridge shall be disassembled in such a manner that determination can be made that the sealant has been applied for the full length of 360° of the joining surfaces at the junction of the "O" ring and case sidewall and that the propellant cavity is free of sealant.

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4.4.7 Functional

4.4.7.1 Ballistics.-The test sample shall be rigidly supported in the fixture. The sample shall be tested within 5 minutes after removal from the conditioning chamber. When this time limit is exceeded, the sample shall be reconditioned in accordance with 4.4.7.1.1 and Table V.

4.4.7.1.1 Temperature conditioning.-Cartridges shall be loaded into acceptable M54 or M52 initiators with end caps to cover the gas ports during the conditioning period. The temperature of the propellant within the cartridges shall be considered as the conditioning temperature and will be determined by means of a thermocoupler attached to the center of the propellant charge of a specimen cartridge assembled in a standard test device similar to the initiator undergoing tests. The conditioning time shall be 1 hour longer than the time necessary to bring the element contained in the specimen cartridge to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-Packing and marking shall be in accordance with Drawings C8864168 and C9214156 except that all markings shall reference Cartridge, Delay, M128 Assembly.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.2 Ignition delay is defined as the time that elapses between the impact of the firing pin on the primer and the beginning of a continuous pressure rise on a pressure time record.

6.3 Storage and Shipping Information

Explosive Hazard Classification - Class I
Storage Compatibility - Group B, E, or N
Department of Transportation (DOT) Hazard - Class C
DOT Marking - "Explosive Power Device - Class C - Handle Carefully, Keep Fire Away"

6.4 Hazard notice.-The cartridge described herein and certain of its components are flammable and/or explosive and consequently present hazards in manufacture, handling, storage and shipment. The contractor should recognize these hazards and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmospheere, rough handling and electrically induced incidents.

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6.5 Supersession data.-This specification includes the requirements of Purchase Description FA-PD-MI-2453, Rev. 1, dated 5 May 1961.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1377-A229

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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